Lecture 1

What is a Recipe?

What is reproducibility?

This all started with the concept of reproducibility

What is reproducibility really?

Ask different people and you'll get suprisingly different answers. Do a straw poll in your group.

Ha! 😈

Looks like the definition of *reproducibility* is not quite *reproducible*

The dark side of reproducibilty (why scientists hate it)

It typically has a negative connotation

Your research sucks!

It is all wrong!

Show me what you did!

My definition of reproducibility

My personal opinion of what **data analysis reproducibility** is:

I want to be able to:

1. Understand **exactly** what the analysis consist of.
2. Access to **all inputs** and **outputs** of the analysis.

Today's scientific practices fail at both levels.

1. Understand what the analysis is

> Understand **exactly** what the analysis consist of.

Why?

- We want to get better what we do.
- We want to be more productive and learn better techniques.
- We want to know if we can improve on the process

2. Data access requirement

“ Access to **all inputs** and **outputs** of the analysis. ”

Why?

It is unrealistic to expect other scientists to process mountains of data, spend weeks and months of work just to verify the correctness of a published process.

All relevant data - even intermediary data should be distributed.

What are recipes?

Our solution to the reproducibility

Recipes

A **recipe** is web based location that is home of:

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1. A **script** that can be run and customized.

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2. A **directory of results** that show all outputs generated by the process.

That's it.

Recipes are universally reusable

If your computer is set up with the properly (see the Biostar Handbook) then

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You can download and run any recipe on your computer!

It will produce the same output that we show.

You are free to experiment, customize, build upon

If you only want to investigate some of the results, download those only and start there.

Bioinformatics Recipes

All content is organized in

- **Projects**

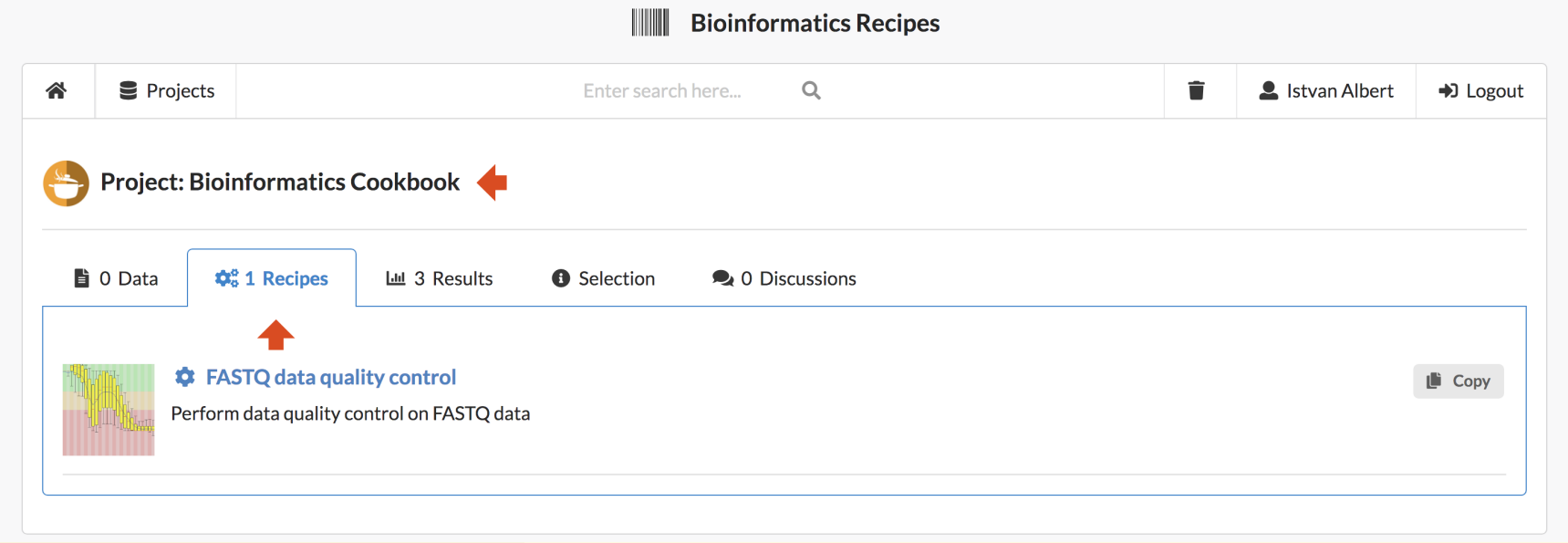
Each **Project** is a collection of:

- **Data**
- **Recipes (scripts)**
- **Results (the result of running a recipe)**

Visit the link in the lecture:
http://www.bioinformatics.recipes

Main Site Interface

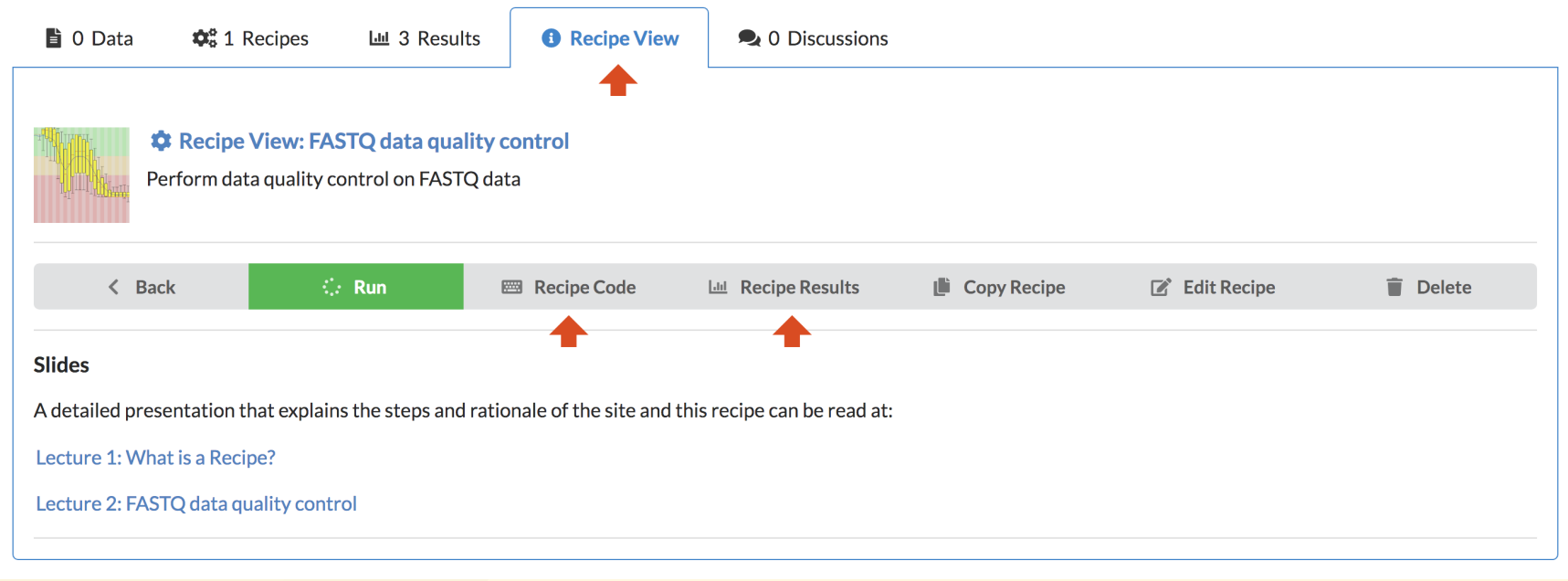
Select the project **Bioinformatics Cookbook**



We selected **Recipes** tab and we see a recipe here named **FASTQ data quality control**

Recipe View

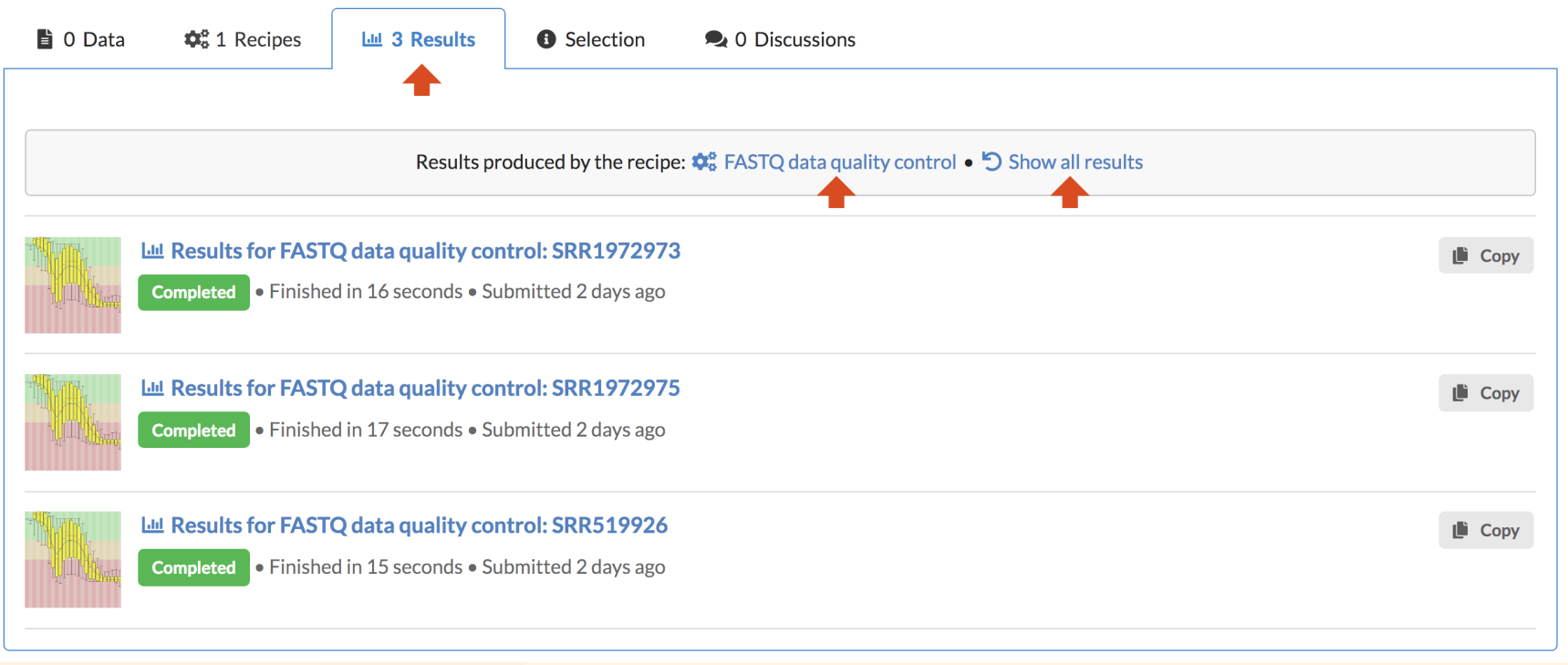
Selecting the recipe shows the **Recipe View**



- **Recipe Code**: what does the recipe consists of
- **Recipe Results**: what does the recipe produce

Recipe Code

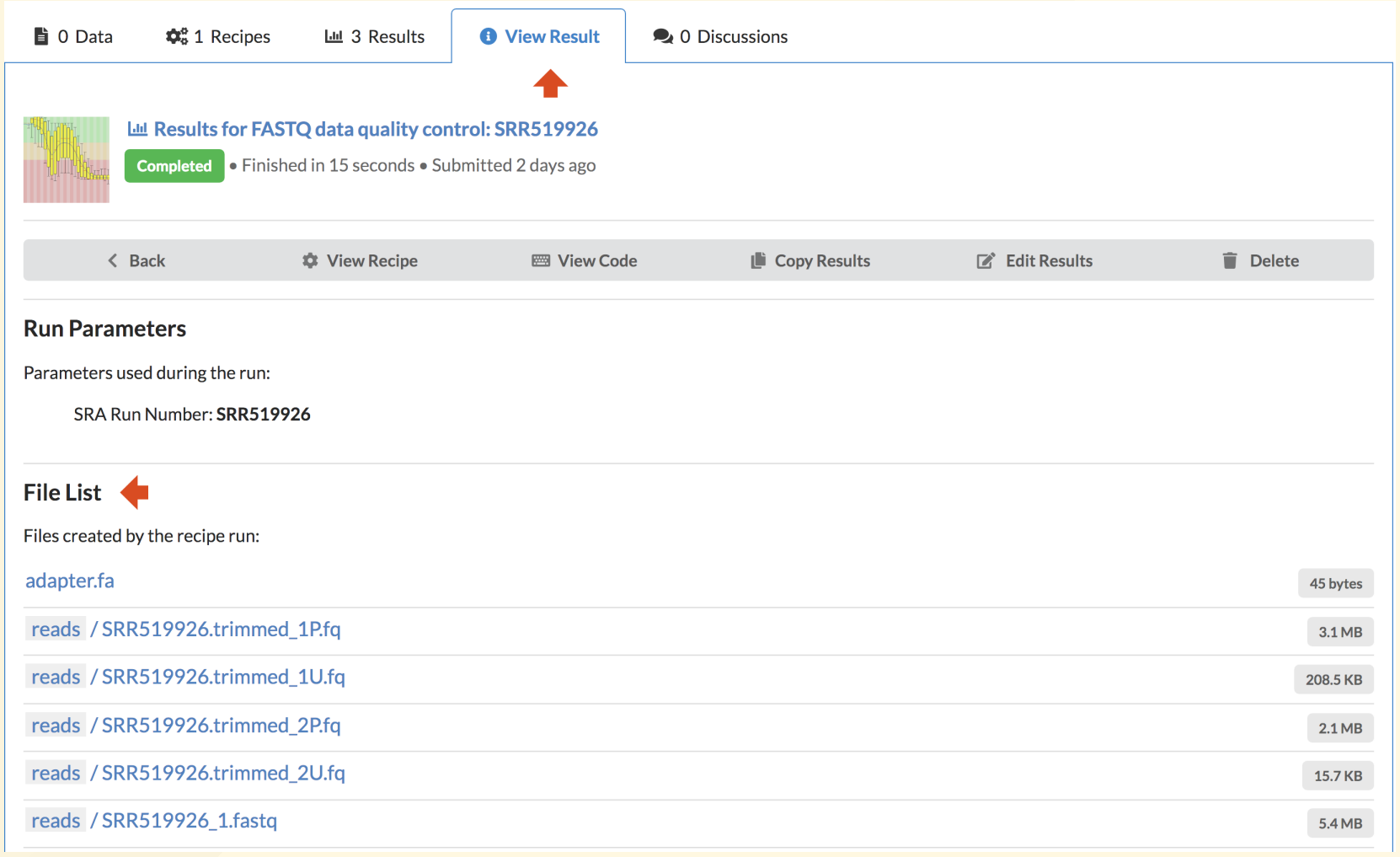
Recipe Result List (runs)



Each result is a product of a "run" of the recipe.

Runs may use different parameters.

Recipe Result View



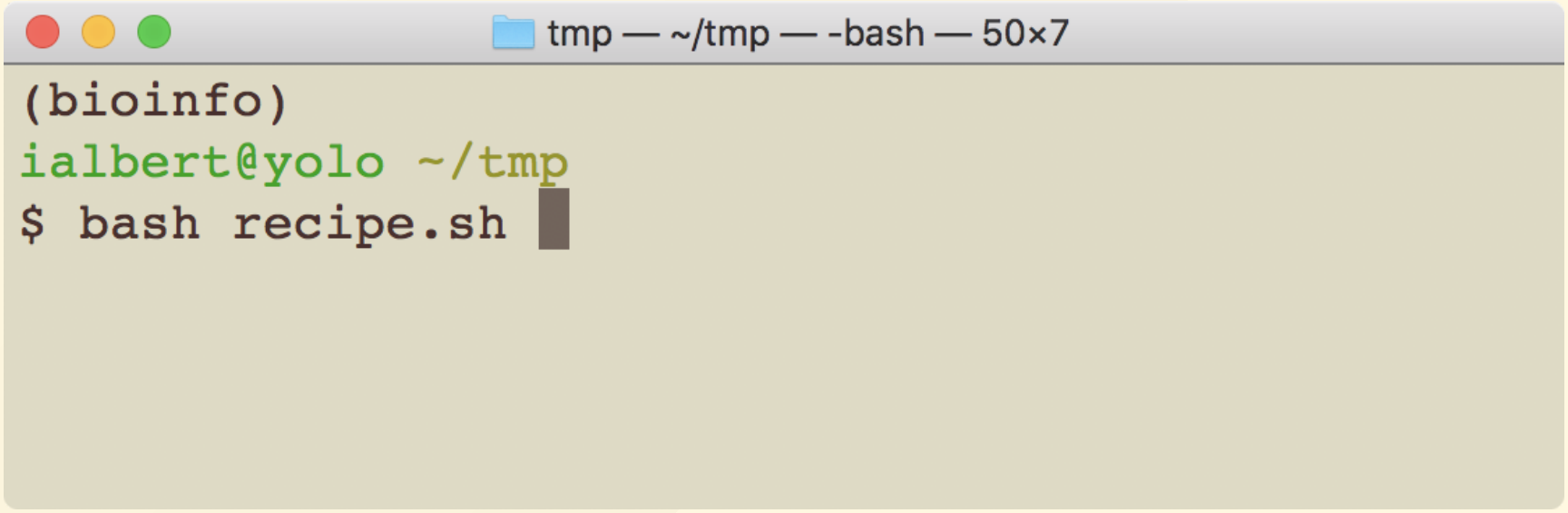
All files created when the recipe was run

Copy the recipe to your computer

```
1  # This recipe downloads sequencing data from SRA
2  # then performs quality filtering and adapter trimming.
3
4  # This is how the recipe gets the SRR
5  # variable filled via the website.
6  SRA={{sra.value}}
7
8  # When used as a script set the SRA number as show below:
9  # SRA=SRR519926
10
11 # Stop the script on errors.
12 set -ue
13
14 # How many sequences to unpack.
15 N=10000
16
17 # Create directory to store the reads in.
18 mkdir -p reads
19
20 # Download 1000 reads from SRA.
21 fastq-dump --split-files -X $N -O reads  $SRA
```

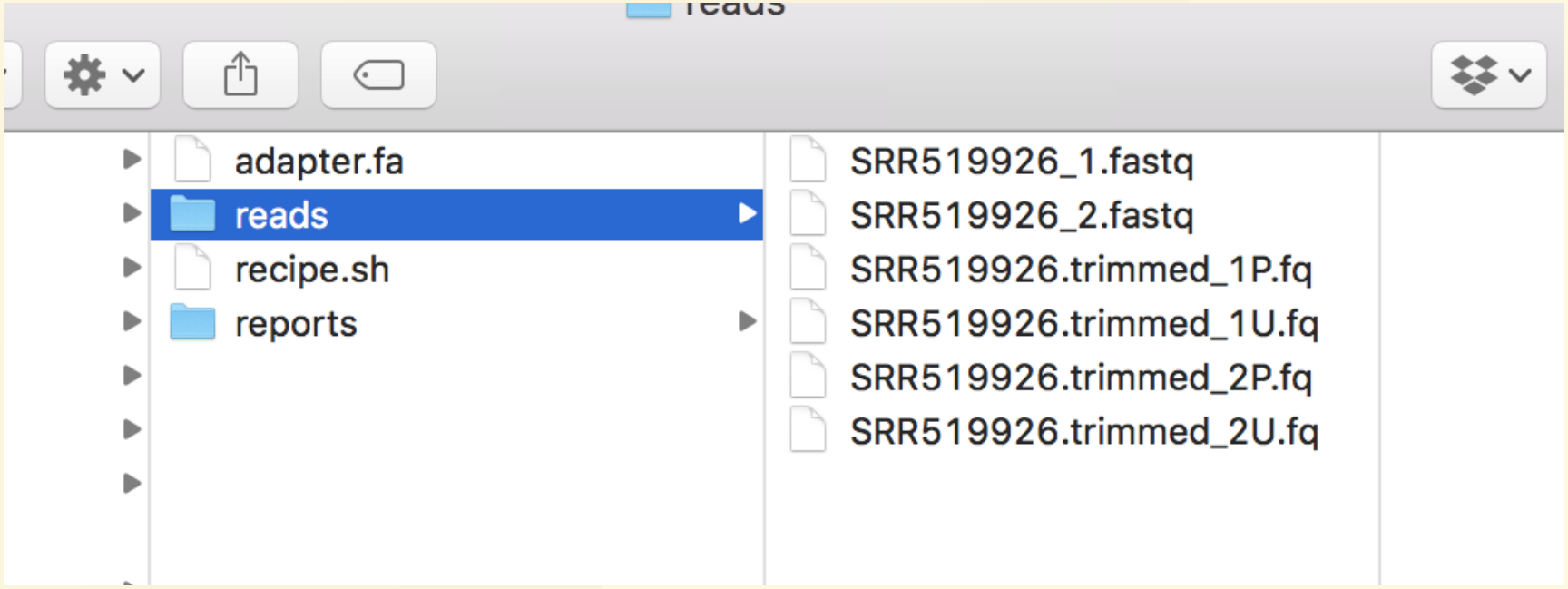
May need small changes: see lines **6** and **9**.

Run the recipe locally



Run the recipe in bash

View the files locally



The exact same files are created locally as on the website.

Reproducibility at its BEST

👍 You can see exactly **how** an analysis works!

👍 You can see precisely **what** an analysis produces!

👍 You can **repeat** the same process on your system!

👍 You can **access** all data generated by the analysis!

That's it!

Bon Appétit! 🍞